

Topic: Hypothesis Testing

Important note for submissions:

- Only handwritten answers are allowed.
 - The hard-copy of solutions should be submitted to me in Wednesday lecture.
 - No email submissions will be entertained.
 - **Submit solutions to ONLY Problems 2, 4, 6, 8, 9, and 10.**
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1. A large manufacturing firm is being charged with discrimination in its hiring practices.
 - (a) What hypotheses are being tested if a jury commits a type I error by finding the firm guilty?
 - (b) What hypotheses are being tested if a jury commits a type II error by finding the firm guilty?
2. A fabric manufacturer believes that the proportion of orders for raw material arriving late is $p = 0.6$. If a random sample of 10 orders shows that 3 or fewer arrived late, the hypothesis that $p = 0.6$ should be rejected in favor of the alternative $p < 0.6$. Use the binomial distribution.
 - (a) Find the probability of committing a type I error if the true proportion is $p = 0.6$.
 - (b) Find the probability of committing a type II error for the alternatives $p = 0.3$, $p = 0.4$, and $p = 0.5$.
3. According to a dietary study, high sodium intake may be related to ulcers, stomach cancer, and migraine headaches. The human requirement for salt is only 220 milligrams per day, which is surpassed in most single servings of ready-to-eat cereals. If a random sample of 20 similar servings of a certain cereal has a mean sodium content of 244 milligrams and a standard deviation of 24.5 milligrams, does this suggest at the 0.05 level of significance that the average sodium content for a single serving of such cereal is greater than 220 milligrams? Assume the distribution of sodium contents to be normal.
4. *Amstat News* (December 2004) lists median salaries for associate professors of statistics at research institutions and at liberal arts and other institutions in the United States. Assume that a sample of 200 associate professors from research institutions has an average salary of \$70,750 per year with a standard deviation of \$6000. Assume also that a sample of 200 associate professors from other types of institutions has an average salary of \$65,200 with a standard deviation of \$5000. Test the hypothesis that the mean salary for associate professors in research institutions is \$2000 higher than for those in other institutions. Use a 0.01 level of significance.
5. A rice manufacturer makes rice packets whose weight is assumed to be a normal random variable with mean 1 kg and standard deviation 10 grams. Samples are drawn hourly and checked. If the production level gets out of sync with a statistical significance more than

- 5%, then the process is stopped and fixed. A sample of 20 rice packets has a mean 992 grams. Should the process be stopped for adjustment? Explain your answer in as much detail as possible. You can use normal distribution, and note that $z_{0.025} = 1.96$.
6. A clinical trial was conducted to determine if a certain type of inoculation has an effect on the incidence of a certain disease. A sample of 1000 rats was kept in a controlled environment for a period of 1 year, and 500 of the rats were given the inoculation. In the group not inoculated, there were 120 incidences of the disease, while 98 of the rats in the inoculated group contracted it. If p_1 is the probability of incidence of the disease in uninoculated rats and p_2 the probability of incidence in inoculated rats, compute a 90% confidence interval for $p_1 - p_2$. Does the inoculation have any impact on the disease incidence rates? Note that $z_{0.05} = 1.645$.
 7. A company has launched a new premium version of its product and claims that the *time it takes customers to adopt the new version* is fairly consistent, with a population variance of 1 years. A random sample of 5 early customers showed the following adoption times (in years): 1.9, 2.4, 3.0, 3.5, and 4.2 years. Construct a 95% confidence interval for the population variance σ^2 of the time-to-adoption and determine whether the company's claim that $\sigma^2 = 1$ is plausible. Assume the population of adoption times is approximately normally distributed. Note that $\chi^2_{0.025,4} = 11.143$ and $\chi^2_{0.975,4} = 0.425$.
 8. A company recently introduced a new customer-service chatbot and wants to determine whether it has improved average customer satisfaction scores.
 - If the analyst commits a Type I error by mistakenly concluding that the chatbot has no effect on customer satisfaction, what null and alternative hypotheses is she testing?
 - If the analyst commits a Type II error by mistakenly concluding that the chatbot does improve customer satisfaction, what null and alternative hypotheses is she testing?
 9. A large consulting firm recently introduced a new “Focus Block” workflow system that encourages employees to schedule uninterrupted work sessions during the week. Management claims that employees using the system will, on average, spend more than 8 hours per week in focused, uninterrupted work. A random sample of 225 employees shows an average of 8.5 hours per week, with a standard deviation of 2.25 hours. Does the sample provide evidence that employees using the new workflow system spend more than 8 hours per week in focused work? Report the P-value and state your conclusion.
 10. State the **null and alternative hypotheses** to be used in testing the following business-related claims:
 - (a) A national retail chain claims that the average monthly in-store footfall is 21.8 thousand customers.
 - (b) A subscription-based app states that no more than 20% of its users cancel their membership in a given month.
 - (c) A delivery company asserts that the average delivery distance per order is 6.2 kilometers.
 - (d) An automobile manufacturer forecasts that at least 70% of next year's vehicle sales will be compact or subcompact.
item A marketing firm claims that the proportion of consumers preferring Brand A is 0.58.
 - (e) A packaged-food company guarantees that the average weight of its premium granola packs is at least 340 grams.